Box distributes over derivability

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source rule \RK{}; normality of Sigma is the standing hypothesis of the canonical-model section.

Editorial note. The label lem:box1 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Normality and the RK rule. Same unit, normality of Sigma and rule \RK{}. From the finitary implication Sigma proves B1 -> (B2 -> ... (Bn -> A)) the RK rule gives the boxed implication.

Lemma 0.1. *If* $\Gamma \vdash [\Sigma]A$ *then* $\Box\Gamma \vdash [\Sigma]\Box A$.

Proof. If $\Gamma \vdash [\Sigma]A$ then there are B_1, ..., $B_k \in \Gamma$ such that $\Sigma \vdash B_1 \to (B_2 \to \cdots (B_n \to A)\cdots)$. Since Σ is normal, by rule RK, $\Sigma \vdash \Box B_1 \to (\Box B_2 \to \cdots (\Box B_n \to \Box A)\cdots)$, where obviously $\Box B_1$, ..., $\Box B_k \in \Box\Gamma$. Hence, by definition, $\Box\Gamma \vdash [\Sigma]\Box A$. □